

Q2 - Attendance Audit System

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a) Data Preparation

So we had 27 photos of handwritten attendance sheets from the semester. Each one has the class number, date, and a list of students who signed in with their names and usernames.

What I Did

1. **Downloaded everything** from the Google Drive link - 27 JPG files
2. **Resized the images** - The originals were huge (like 2600x3300 pixels) which caused problems. I shrunk them down to 1200x1200 using PIL:

```
python from PIL import Image img = Image.open(path)
img.thumbnail((1200, 1200)) img.save(output)
```
3. **Didn't need to do much else** - The photos were pretty clear. No need for fancy preprocessing like binarization or noise removal. The handwriting was readable enough.

Why I Kept It Simple

Honestly, the sheets were consistent. Same format every time - class number and date at the top, numbered list of names below. The vision model could read them fine without me doing a bunch of image processing first.

b) How I Built This

The Approach

I used a **vision-capable LLM** (basically a model that can look at images and understand text) to read each attendance sheet. This is a pre-trained model - I didn't train anything myself.

Why This Made Sense

- Handwriting is hard for traditional OCR (Tesseract struggles with it)
- Training my own model would need labeled data I don't have
- The VLM can understand context - it knows "Class 15" and "Oct 7" are header info, not student names
- Way faster than typing everything by hand

The Process

1. Fed each resized image to the vision model
2. Extracted: class number, date, and student count
3. Stored everything in a Python script with the data hardcoded
4. Ran stats on it (median, min, max, correlations)

Checking My Work

- Made sure class numbers matched up with dates sequentially
 - Cross-checked against filenames (some had dates in them like "Nov18-...")
 - Spot-checked a handful manually to make sure counts were right
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c) The Results

c.a) Classes and Dates

27 classes total, running from August 19 to November 20, 2025.

Here's the full list:

#	Date	#	Date
1	Aug 19	15	Oct 7
2	Aug 21	16	Oct 14
3	Aug 26	17	Oct 16
4	Aug 28	18	Oct 21
5	Sep 2	19	Oct 23
6	Sep 4	20	Oct 28
7	Sep 9	21	Oct 30
8	Sep 11	22	Nov 4
9	Sep 16	23	Nov 6
10	Sep 18	24	Nov 11
11	Sep 23	25	Nov 13
12	Sep 25	26	Nov 18
13	Sep 30	27	Nov 20
14	Oct 2		

Classes were Tuesdays and Thursdays.

c.b) Median Attendance

35 students

Average was about the same (34.7). Attendance bounced around quite a bit though - anywhere from 16 to 49 depending on the day.

c.c) Highest and Lowest

Lowest: 16 students - Class 27, November 20th

This was right before Thanksgiving break. Makes sense that people bailed early. Class 25 (Nov 13) was also low at 19.

Highest: 49 students - Class 2, August 21st

Second class of the semester. Everyone's still motivated at that point. The first few weeks all had 40+ students.

c.d) Quiz Days vs Regular Days

The exam asked about correlation with evaluation dates. Here's what I found:

Event	Date	Attendance
Quiz 2	Oct 7	45
Quiz 3	Nov 11	41
Paper presentations	Nov 18	34

Average on quiz/eval days: 40 students

Overall average: 34.7 students

So yeah, **there's a correlation**. About 5 more people showed up when something was graded. Not shocking - people skip regular lectures but show up when it counts.

When was attendance highest overall?

- Start of semester (Classes 1-3 had 44-49 students)
 - Quiz 2 on Oct 7 had 45
 - After that it trends down, bottoming out at the end of November
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d) What I'd Do With More Time

If I had another week, here's what would make this better:

1. **Train a custom OCR model** on these specific handwriting styles. Would take some manual labeling but would be more accurate than the general vision model.
2. **Build an actual pipeline** - right now it's pretty manual. Would be nice to have something where you just drop in an image and it spits out the data automatically.

3. **Cross-reference with class roster** - check if the names I extracted actually match enrolled students. Catch any weird spellings or people signing in who shouldn't be there.
 4. **Predict future attendance** - with this much data you could probably build a decent model. Factor in day of week, proximity to breaks, whether there's a quiz, etc.
 5. **Make a dashboard** - something visual where you can see trends over time. Would be useful for an instructor to spot patterns.
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Code

Everything's in `./code/attendance_analysis.py`

Run it with: `bash python3 attendance_analysis.py`

It'll print out the stats and save CSV/JSON files with all the data.